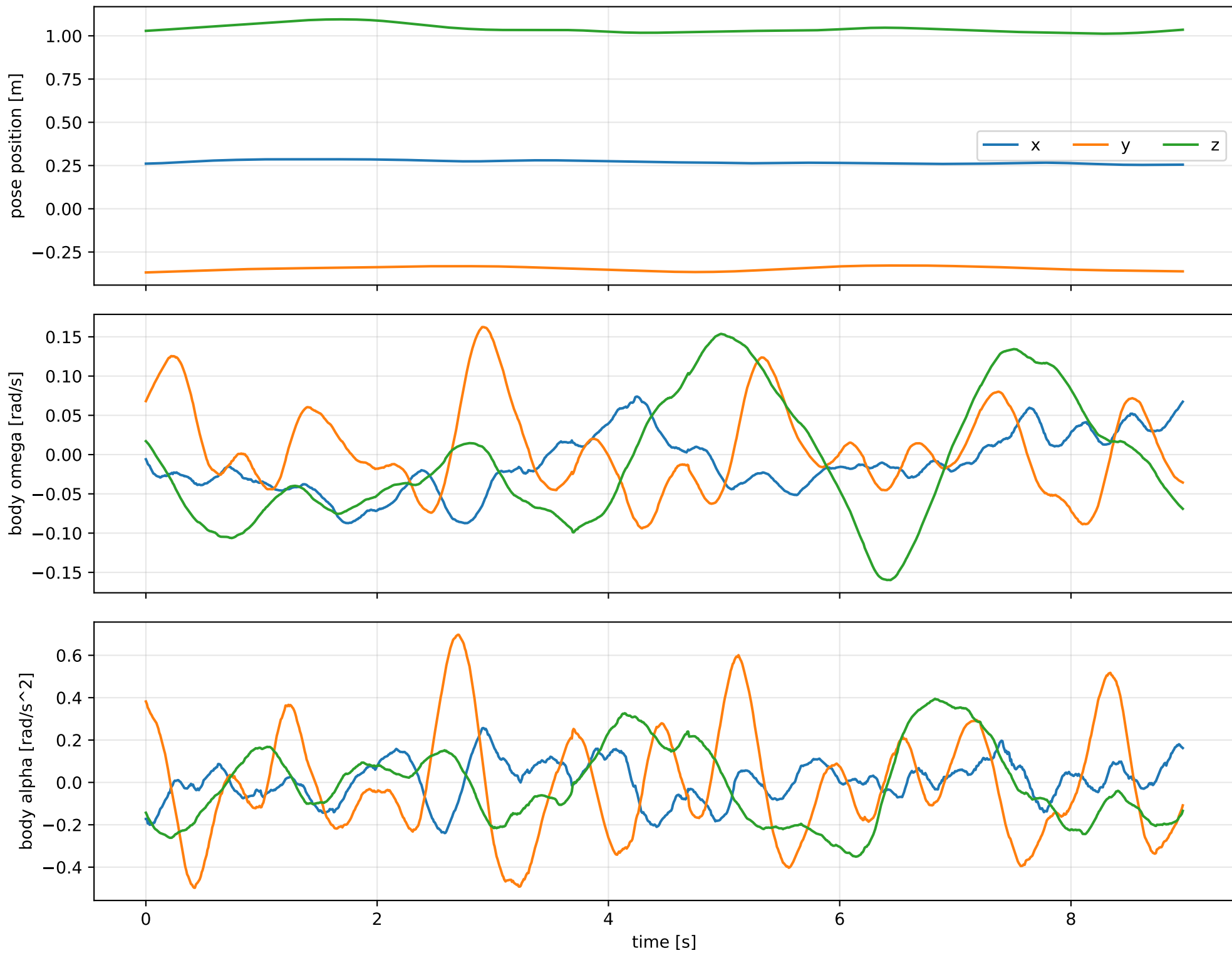
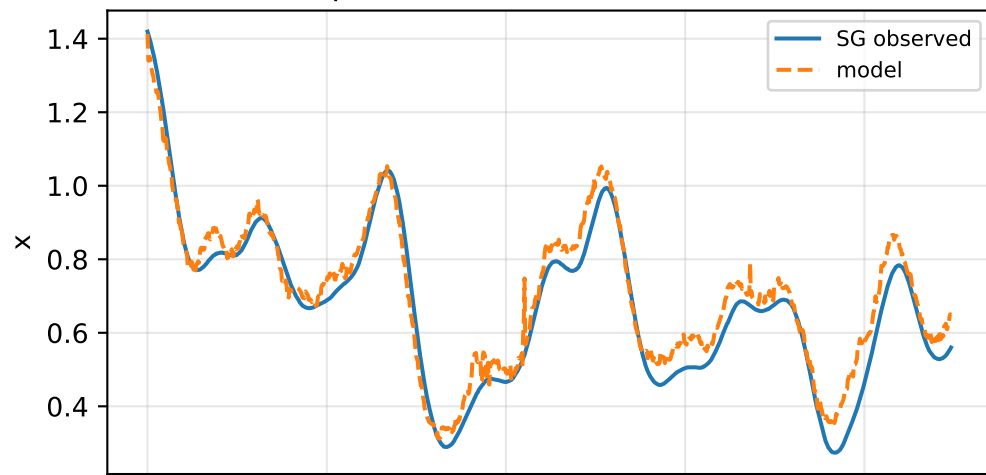


gimbal_command_replay: SG trajectory and derived motion

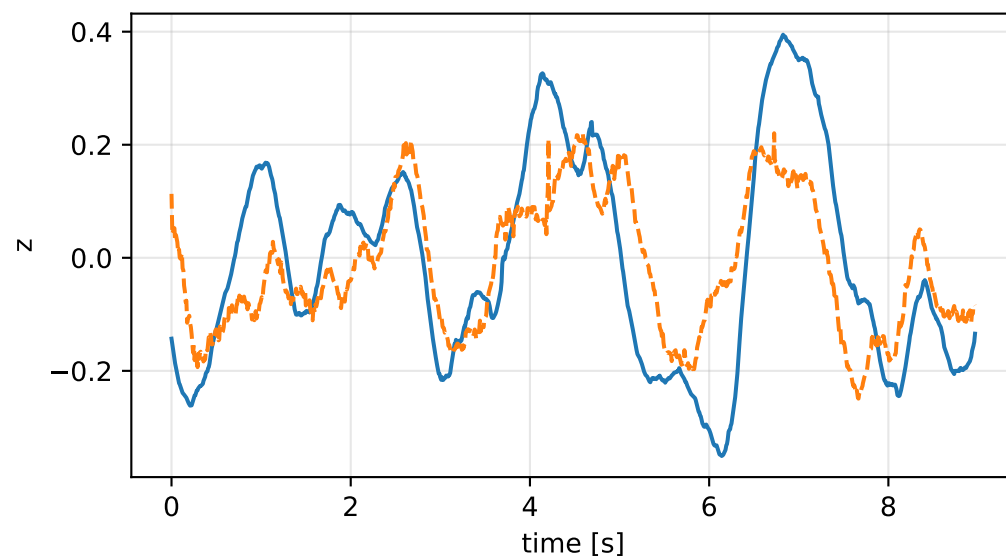
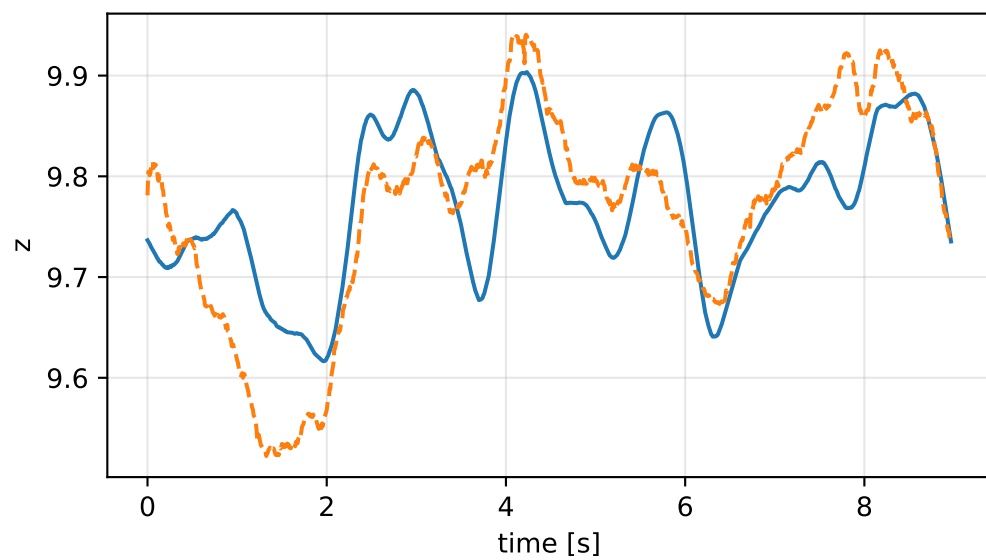
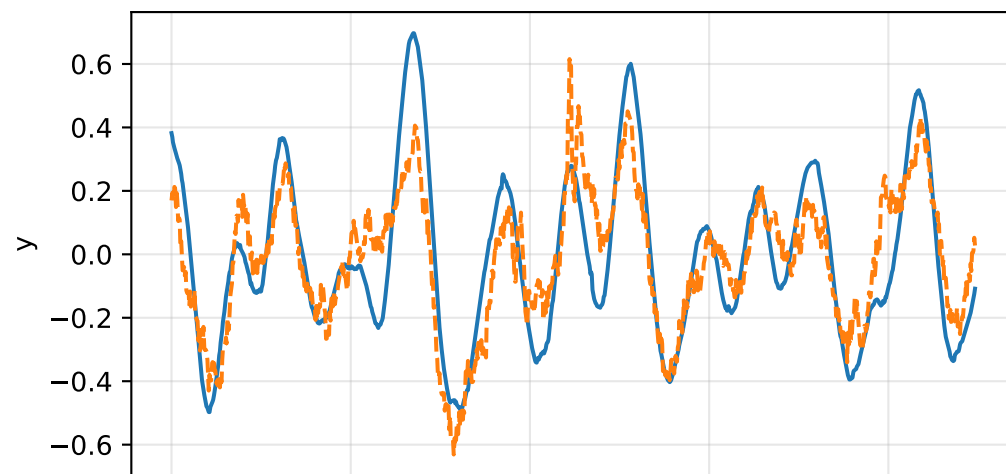
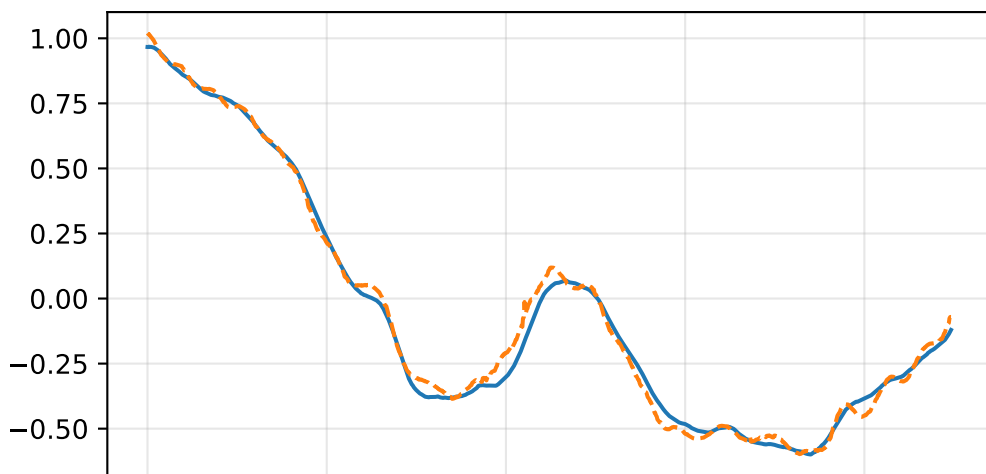
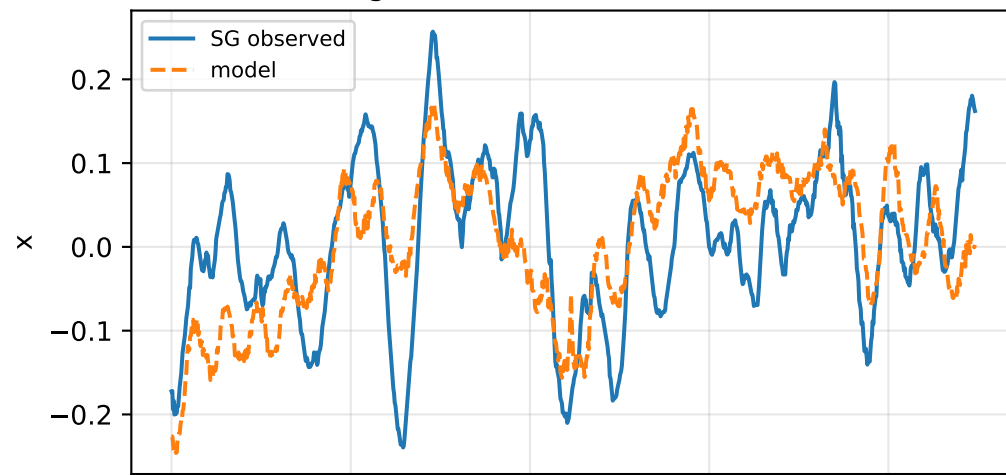


gimbal_command_replay: acceleration residual objective

specific acceleration [m/s²]

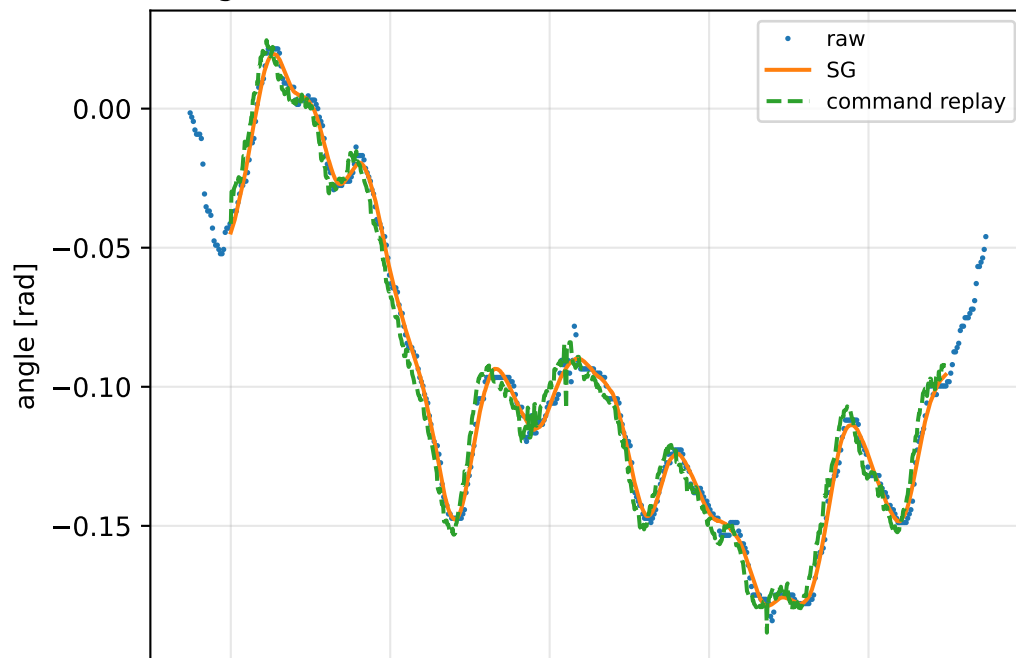


angular acceleration [rad/s²]

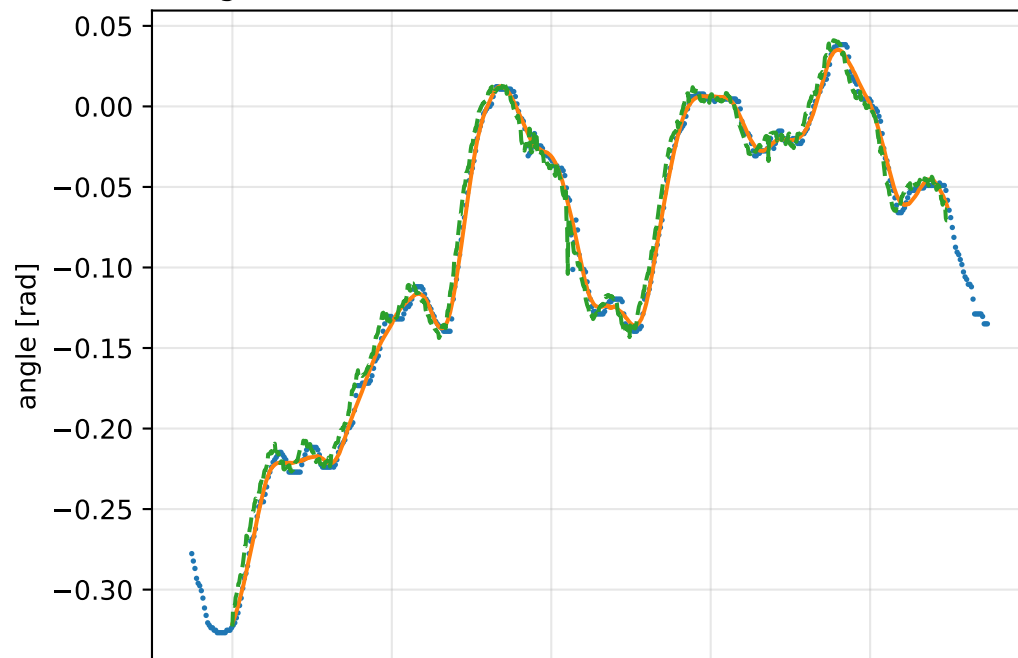


gimbal_command_replay: gimbal measurement audit (objective=command_replay)

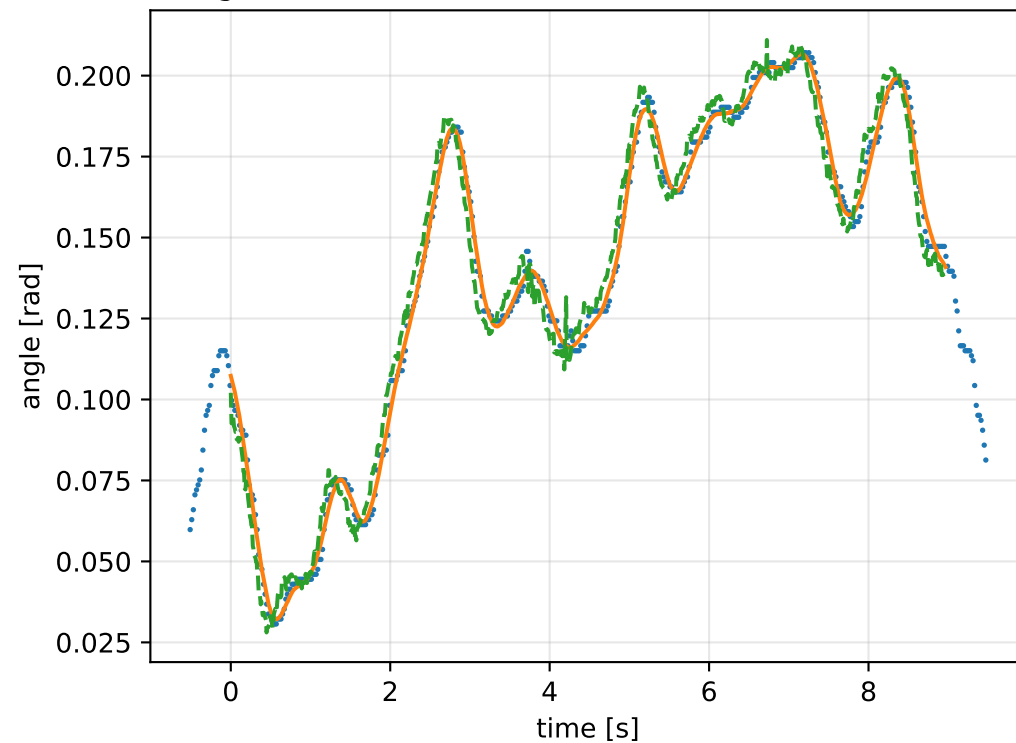
gimbal 1: RMSE=0.006345, max=0.01494 rad



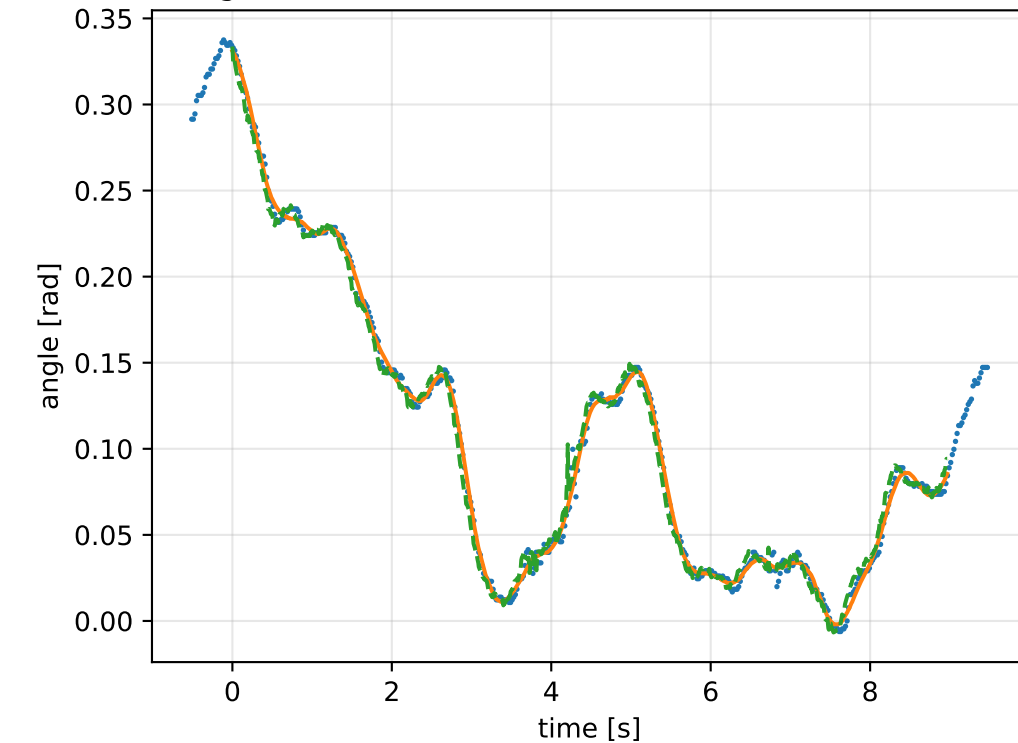
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

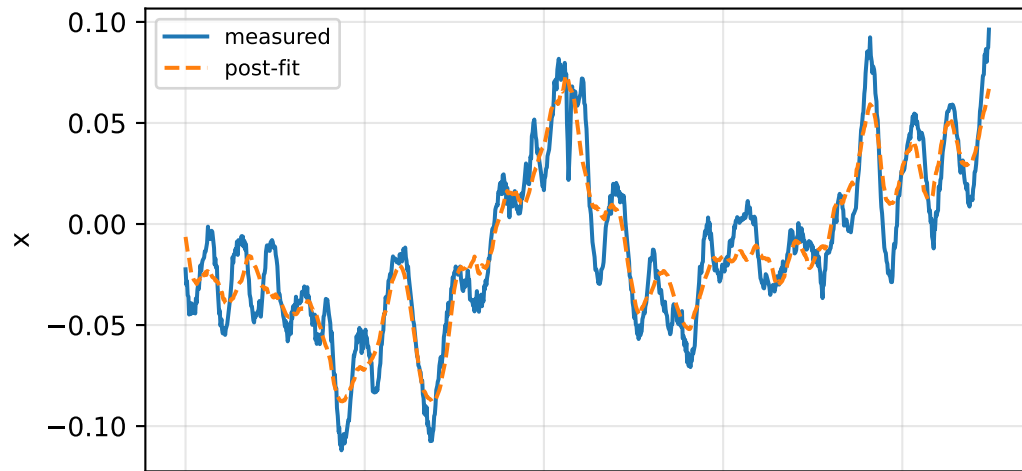


gimbal 4: RMSE=0.006321, max=0.03642 rad

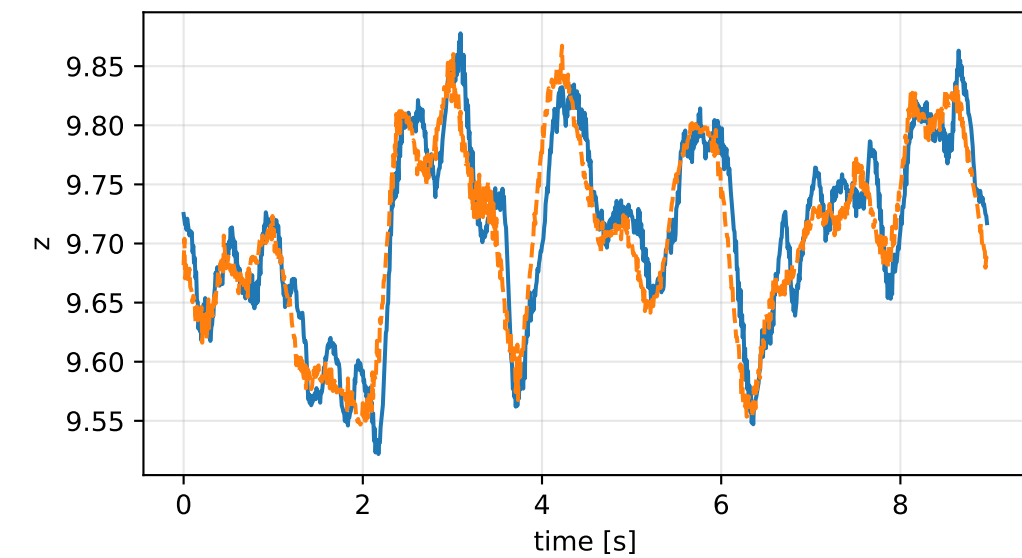
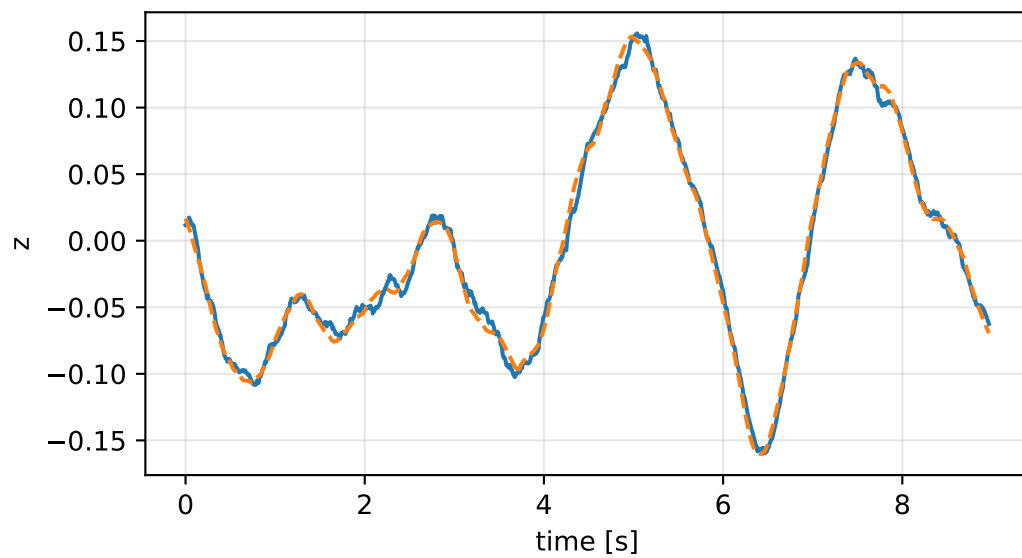
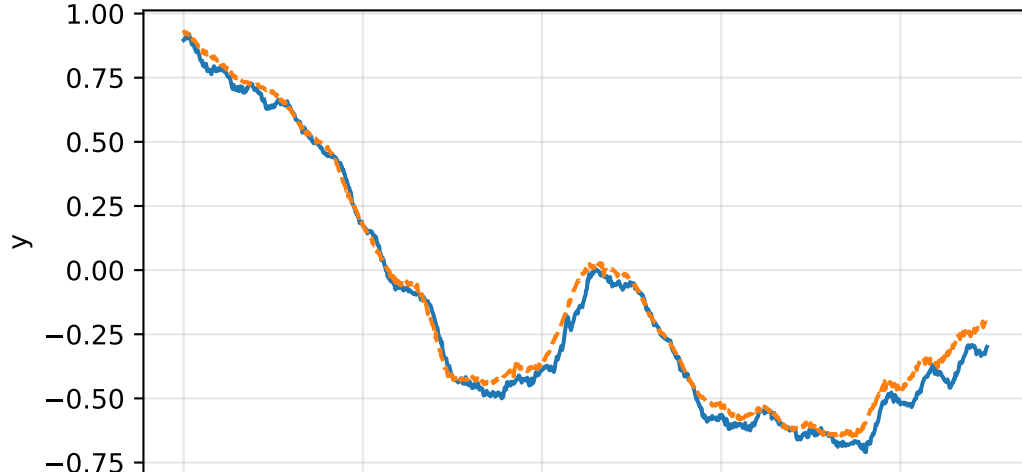
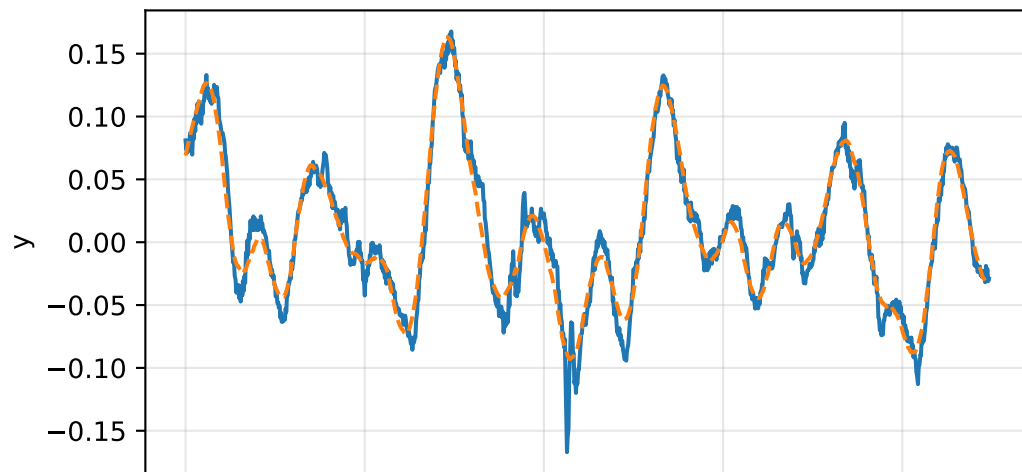
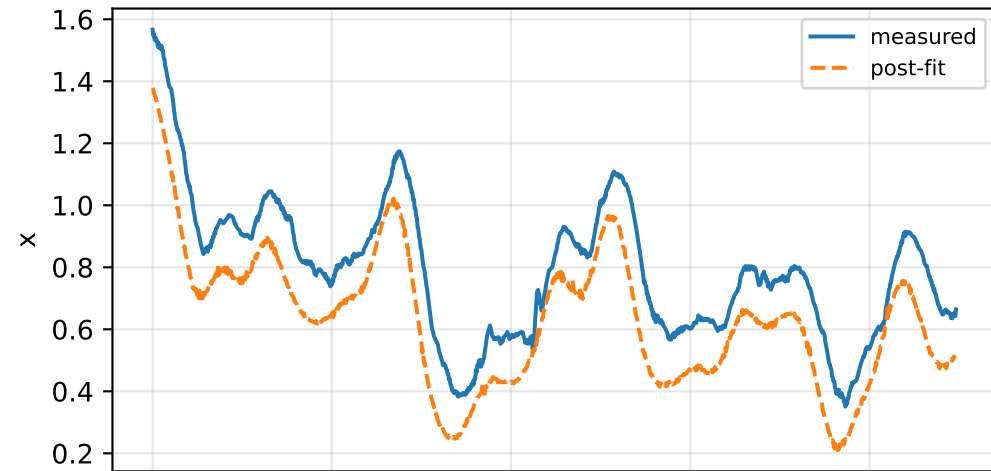


gimbal_command_replay: IMU comparison (diagnostic only)

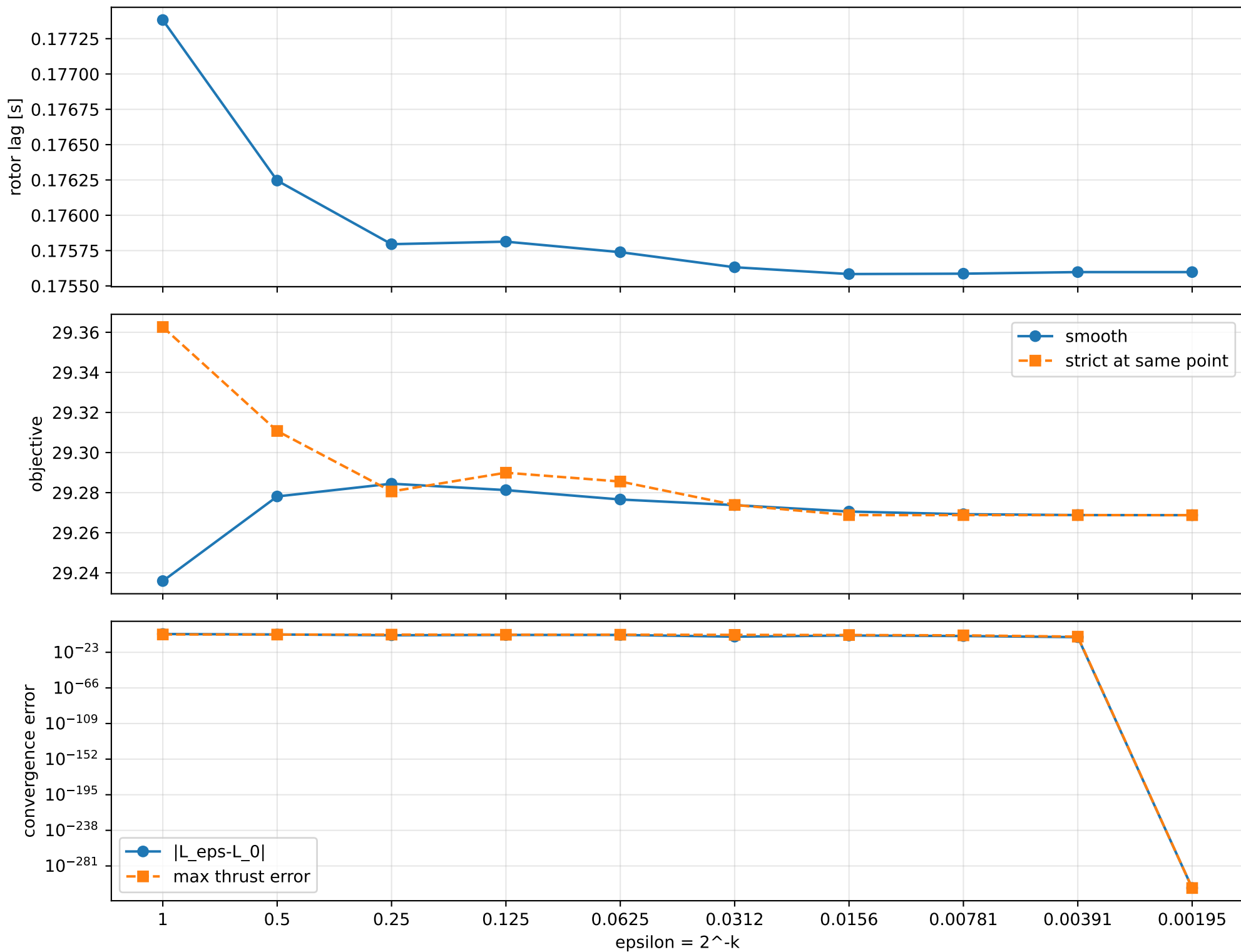
gyro [rad/s]



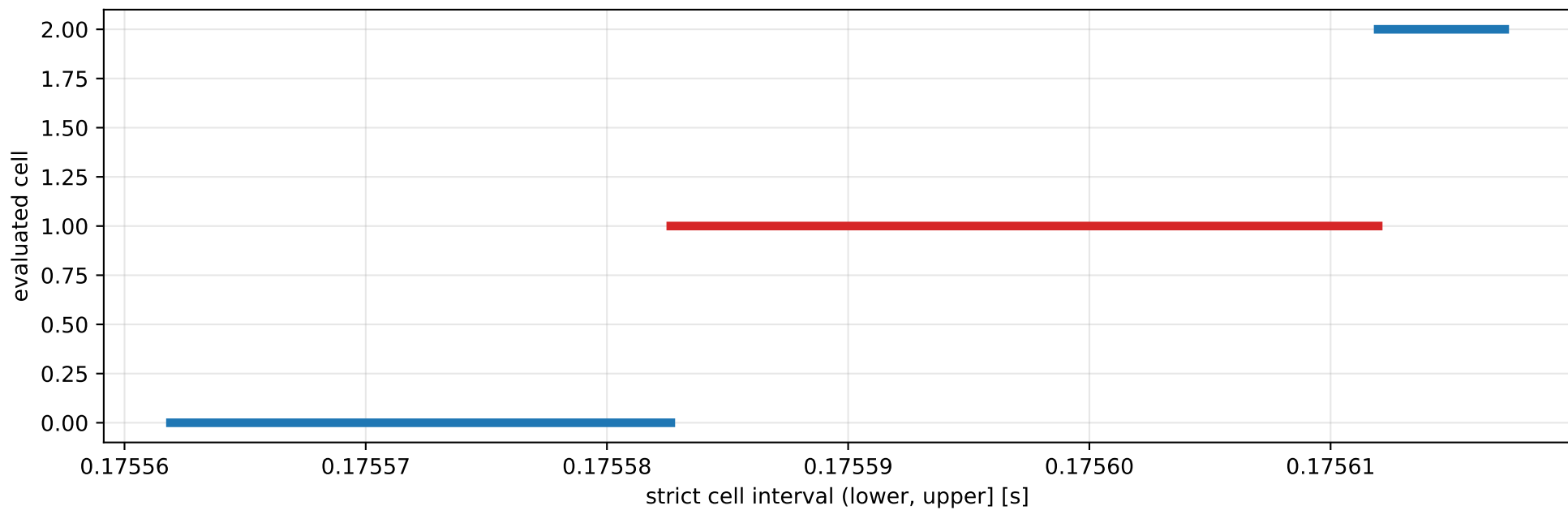
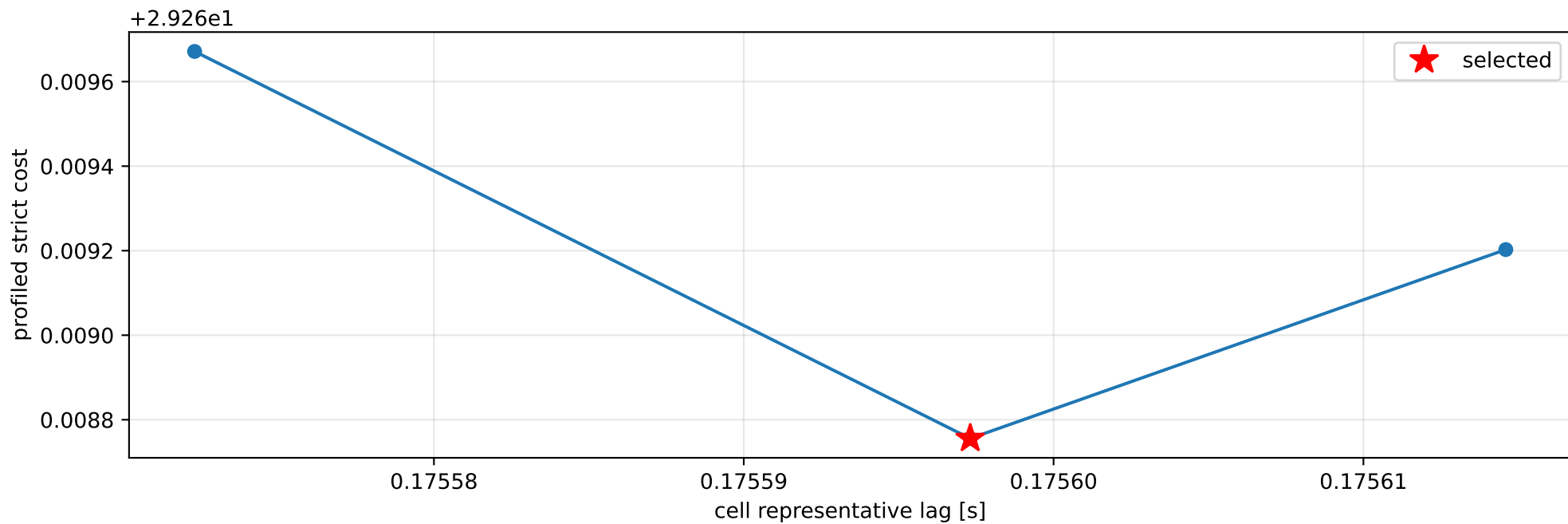
specific force [m/s^2]



gimbal_command_replay: smooth-to-strict continuation



gimbal_command_replay: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597936 s; data boundary=False

gimbal_command_replay: parameter estimate

Case: gimbal_command_replay

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.20958754

inertia [kg m²]:

```
[[ 1.26460785 -0.01656331  0.09182049]
 [-0.01656331  0.56234672 -0.00487402]
 [ 0.09182049 -0.00487402  0.73395782]]
```

CoG body [m]: [-0.01908431 -0.0597558 -0.00550874]

force effectiveness: [2.31909011 2.34190275 1.13518725 1.4369801]

scale-free J/m [m²]:

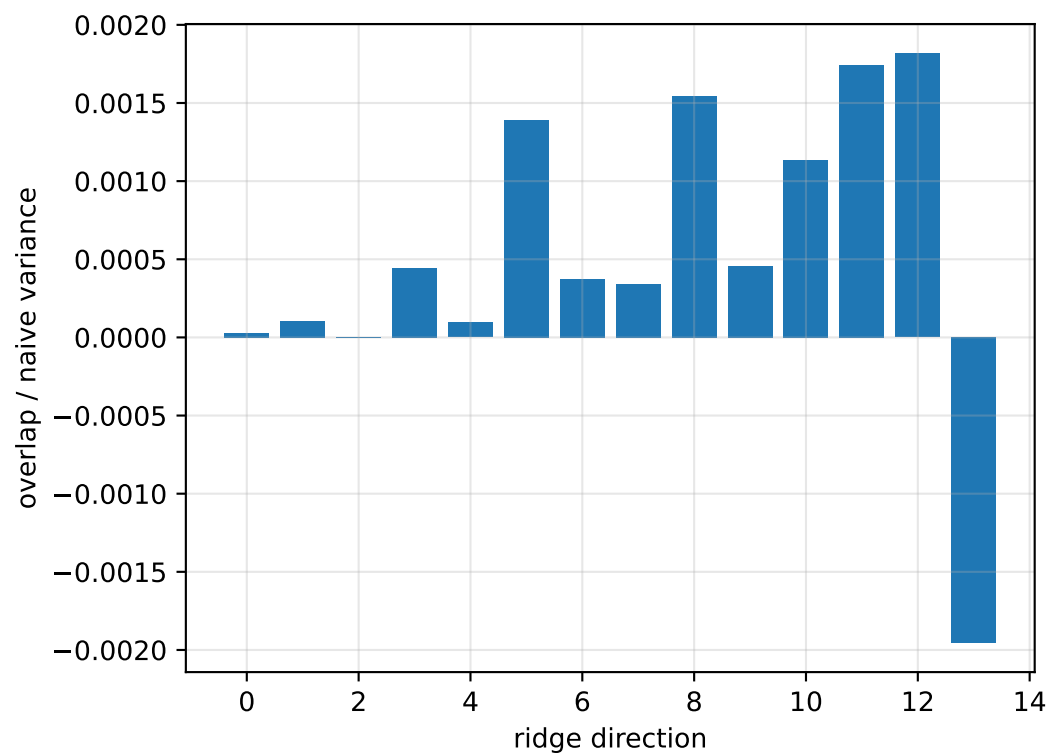
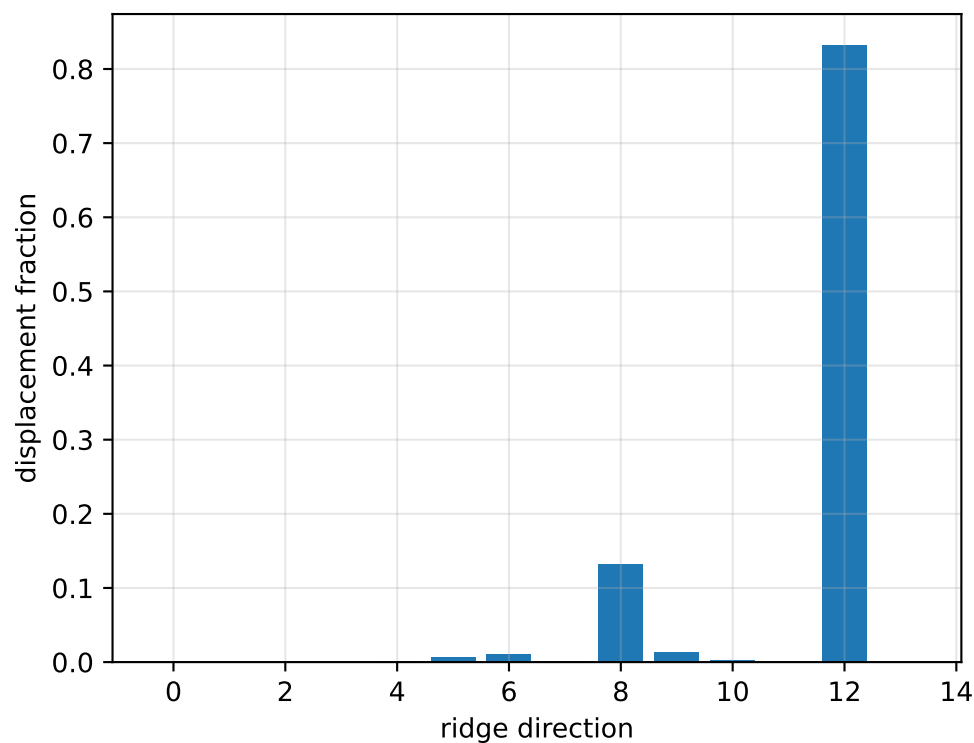
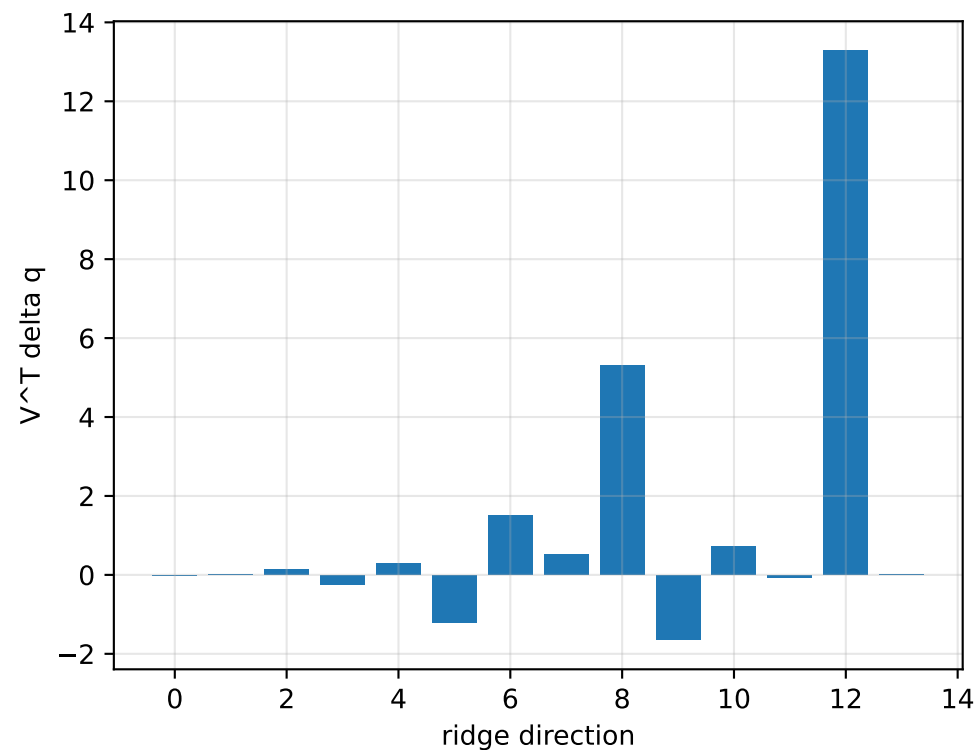
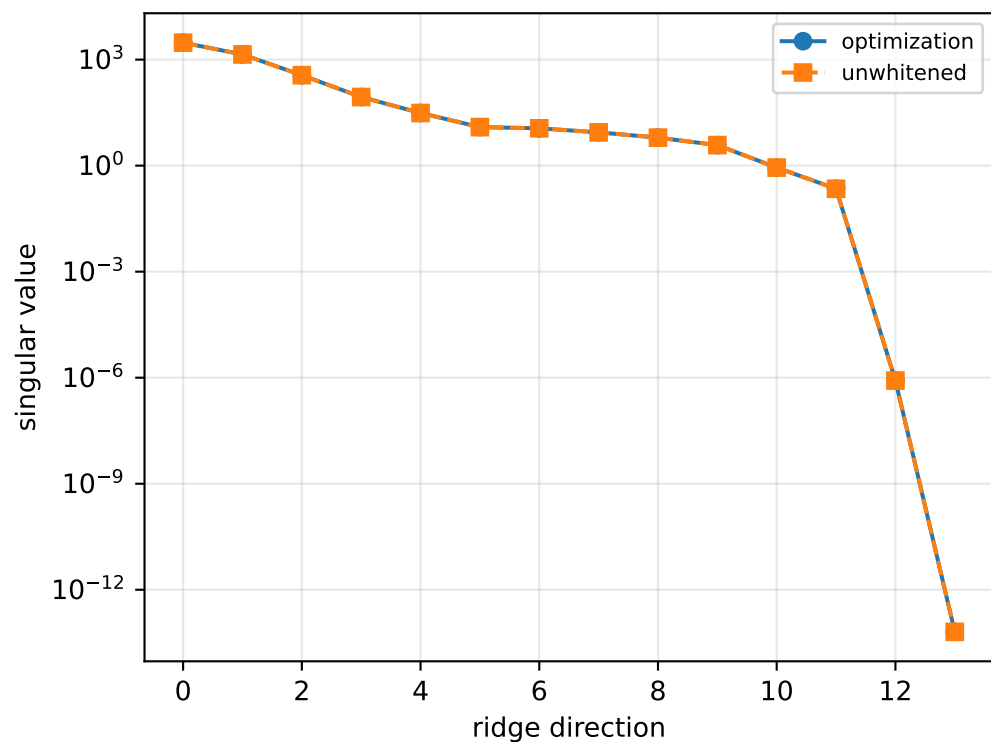
```
[[ 0.24274625 -0.00317939  0.01762529]
 [-0.00317939  0.10794458 -0.00093559]
 [ 0.01762529 -0.00093559  0.14088598]]
```

scale-free f/m: [0.4451581 0.44953708 0.21790348 0.27583376]

rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

gimbal_command_replay: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=7.68\text{e-}14$)



gimbal_command_replay: optimization vs reference covariance



gimbal_command_replay: wrench / closure (force max 1.54e-14, torque max 1.11e-16)

